Supplemental Material: Complex-contour regularization of second-order point-particle sources

S1. CONVENTIONS AND THE LINEAR-SOLVE PROPERTY

Modes are $\Phi = \sum_\ell \phi_\ell(r) Y_{\ell 0}(\theta)$, with $[f]_\ell = \sqrt{\pi(2\ell+1)}\int_{-1}^{1} f\, P_\ell(\mu)\, d\mu$. The first-order mode source is $\kappa_\ell\, \delta(r-r_0)$, with $\kappa_\ell = -(q/r_0^2)\sqrt{4\pi(2\ell+1)}$. The ($\kappa_\ell = 1$) Coulomb mode is $G_\ell = -r_0^2\, r_<^\ell / [(2\ell+1) r_>^{\ell+1}]$.

We use $H(z) = 1 + \frac{i}{\pi}\log(z - r_0)$ with the principal logarithm, and real-axis values are boundary values from above. Logarithms of the form $\log[c(z-z_*)]$ are rewritten as $\ln|c| + \log(z-z_*) - i\pi[c<0]$. This is the same function in the upper half-plane, and its principal value on the axis is the value from above. Without this rewrite, Mathematica's $\log(r_0 - r)$ at real $r > r_0$ lies on the wrong side of the cut.

Linear-solve property. $\mathcal{L}_\ell$ has real coefficients that are analytic away from $z = 0$. The solution continued along the contour therefore equals the boundary value from above of the solution in the region between the contour and the real segment. Taking real parts commutes with $\mathcal{L}_\ell$. So $\mathrm{Re}\,\tilde{\psi}(r + i0)$ solves $\mathcal{L}_\ell \psi = \mathrm{Re}\,\tilde{s}(r + i0)$ with the same real data at a_L. Because $(\Delta r + i0)^{-n} = \mathrm{FP}\,\Delta r^{-n} + i\pi(-1)^n \delta^{(n-1)}(\Delta r)/(n-1)!$, the real part is the Hadamard finite part.

S2. CONSISTENCY OF THE CONTINUED SINGULAR MODE

(i) Real part. Any solution ψ of $\mathcal{L}_\ell \psi = \delta_{\rm ext}$ satisfies $\mathrm{Re}\,\psi(r + i0) - G_\ell = \alpha r^\ell + \beta r^{-\ell-1}$, because both sides solve the same real equation with the same δ source. The singular structure at r_0 does not fix α and β; the 3D puncture does.

`DSolve`'s particular solution with constants set to zero gives the following ($\ell \le 12$ checked):

- $\ell = 0$: an anti-physical Green's function, $(\alpha, \beta) = (r_0, r_0^2)$;
- $\ell = 1$: the Coulomb mode;
- $\ell \ge 2$: a one-sided Green's function that vanishes for $r < r_0$, $\alpha = r_0^{1-\ell}/(2\ell+1)$.

Summed in 3D, these offsets give $-q/\rho + q\, r\cos\theta / r_0^2 - q/r$ for $r < r_0$, and diverge for $r > r_0$. If such a mode is used with 3D SS and SR, the computed second-order source is $Q[\Phi_1 - \Delta, \Phi_1 - \Delta]$ instead of $Q[\Phi_1, \Phi_1]$.

(ii) Imaginary part. Replacing $\Theta \to H$ in the Coulomb mode does not give a solution of $\mathcal{L}_\ell \psi = \delta_{\rm ext}$, because $\delta_{\rm ext}$ is not annihilated by smooth factors that vanish at r_0. For $\ell = 0$, $\mathcal{L}_0[-r_0 + (r_0 - r_0^2/z)H] = \delta_{\rm ext} - \frac{i}{\pi} z^{-1}$. The residual field then acquires an $O(1)$ shadow, $r/(2\pi)$. The fix is to add $\frac{i}{2\pi} z$. Equivalently, use $\psi_p - D_\ell$, where D_ℓ is the homogeneous offset found in (i).

S3. SHADOW GROWTH WITH TRUNCATED PUNCTURES

With $\Phi^{\rm P} = q/\rho + cq\rho^n$ used consistently, the real-part mismatch vanishes identically. The remaining contamination from products of continued modes is linear in c. At fixed $\ell'_{\rm max}$ it falls by about 0.07–0.08 per unit of n, but each coupled mode's contribution grows with ℓ'. Near $\ell' = 30$ the growth per mode is 1.13 and 1.19 for $R = \frac{1}{2}$, and 1.06 and 1.09 for $R = \frac{1}{4}$, for the $Q = AB$ and $Q = \nabla A \cdot \nabla B$ toys respectively. These are close to $1 + R/r_0$ times a slowly varying power.

For fixed R the total error therefore has a minimum in $\ell'_{\rm max}$. Shrinking the worldtube as modes are added, $R = r_0/\ell'_{\rm max}$, keeps $(1 + R/r_0)^{\ell'_{\rm max}}$ bounded, and the error then falls monotonically (Fig. S1). For $Q = \frac{1}{2}\partial_z^2(AB)$ the minimum lies near $\ell'_{\rm max} = 12$–16 (main text, Fig. 4). The hybrid split removes the contamination entirely.

S4. ROUTES FOR CONTINUING SS

In the toy the SS modes are closed form, so they can be continued directly. In GR they are available only from numerical angular quadrature. We compared three ways of putting them on the contour, treating the toy data as if numerical (Fig. S2).

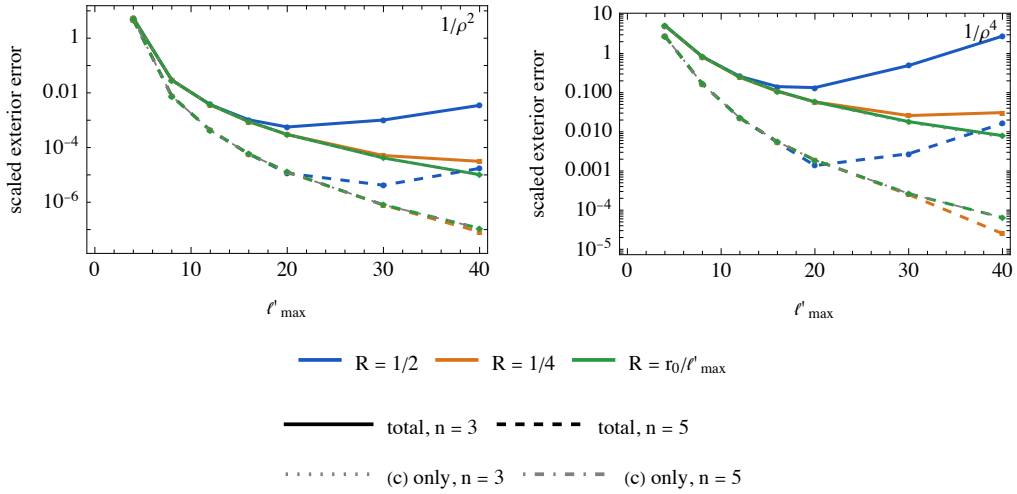


FIG. S1. Products of continued modes with a truncated puncture, for $Q = AB$ (left) and $Q = \nabla A \cdot \nabla B$ (right). Exterior error against $\ell'_{\max}$ for $n = 3$ (solid) and $n = 5$ (dashed): fixed worldtubes $R = \frac{1}{2}, \frac{1}{4}$, and a shrinking worldtube $R = r_0/\ell'_{\max}$. Dotted: mode-truncation error.

- *Route 2 (hybrid).* Subtract the analytic singular terms and Cauchy-continue the real-axis remainder. Subtracting only $A_\ell/\Delta r^2$, or $A_\ell/\Delta r^2 + B_\ell/\Delta r$, fails (0.4 and 0.3 at grid spacing $h = 0.005$), because the remaining $\ln|\Delta r|$ cannot be interpolated. Including $C_\ell \ln|\Delta r|$ gives $O(h^3)$ (3×10^{-7}), and the next two log terms give $O(h^5)$ (3×10^{-12}). However, 10^{-8} relative noise on the SS samples produces an error of 7×10^{-4}: subtracting $A_\ell/\Delta r^2$ amplifies the noise by 10^4–10^5.

- *Radial factor.* Continue $F_\ell = (r - r_0)^2 \mathrm{SS}_\ell$, which is bounded, and divide by $(z - r_0)^2$ on the contour. The shadow of F at r_0 must be pinned to remove spurious δ' and δ terms. Without pinning the error is 23, matching the predicted $-\pi G_0 \delta' + \pi G_1 \delta$ to 10^{-14}. With exact pinning constants the error is $O(h^3)$. With constants taken from the data, G_1 is wrong by $O(h \ln h)$, because of the $\Delta r^2 \ln|\Delta r|$ term in F, and the method is only first order.

- *Route 1 (complex-r quadrature).* Evaluate $\mathrm{SS}_\ell(z) = \oint Q[\Phi^{\mathrm{P}}, \Phi^{\mathrm{P}}](z, \theta)\, Y_{\ell 0}\, d\Omega$ directly at the contour nodes. The integrand is smooth except near the crossing of $|z| = r_0$, where the angular path is deformed. The result agrees with the closed-form continuation to 1.8×10^{-14} on the contour and 1.7×10^{-12} in the exterior, at a cost similar to real-axis quadrature near r_0. No singular quantity is subtracted.

Route 1 also works for SR Part 1 (10^{-13}), and for the full SR with an exact puncture (4×10^{-14}). For pieces whose modes have kinks at r_0, the angular integral at complex r gives a single analytic branch. The inside and outside branches of the *undifferentiated* products must therefore be blended with H before ∂_z^2 is applied. Blending the finished source instead fails (errors 3–900).

Summing continued residual modes at complex r, $\sum_{\ell'} \tilde{\phi}^{\mathrm{res}}_{\ell'}(z) Y_{\ell' 0}(\theta)$, is asymptotically divergent for truncated punctures: terms grow by 1.05–1.07 per ℓ' beyond $\ell' \approx 30$. This is why SR Part 2 is formed on the real axis.

S5. CONVERGENCE OF THE HYBRID SPLIT

With $\Phi^{\mathrm{P}} = q/\rho + cq\rho^n$, the residual $\Phi^{\mathrm{R}} - cq\rho^n$ is C^{n-1} at $\mathbf{x}_0$. Its coupled-mode contributions fall as $\ell'^{-(n+3)}$, so the truncation error falls as $\ell'^{-(n+2)}_{\max}$. The fitted slopes are 2.99, 4.95, 6.91 and 8.87 for $n = 1$, 3, 5 and 7 (Fig. S3). Inside r_0 the error is largest next to the worldtube. Outside, it grows towards $r = 10$, because the Cauchy data are set at $r = 1$.

Fitting the large-ℓ' tail with two terms of the $\Lambda_{\ell,s,n}$ functions of Refs. [1, 2] gains a factor of 38–113 at $\ell'_{\max} = 20$ and up to 2000 at $\ell'_{\max} = 30$. At $\ell'_{\max} = 12$ the two-term fit makes the error worse; a one-term fit gains a factor of about 1.5 there. The hybrid split adds no error of its own: the Cauchy step contributes below 10^{-14}, and the split reproduces the mode-truncation error to better than 10^{-6} relative.

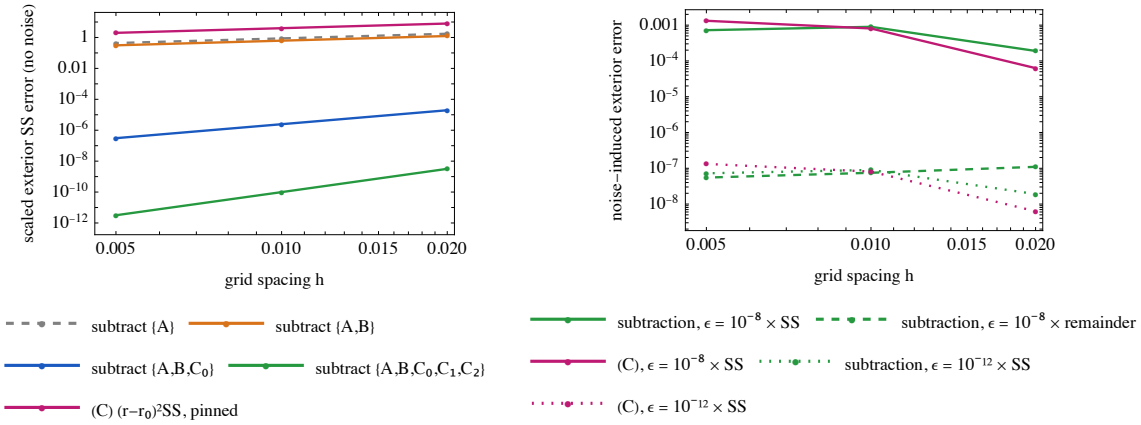


FIG. S2. Continuing SS from data treated as numerical. Left: error against grid spacing for the subtraction variants and the pinned radial factor. Right: noise propagation.

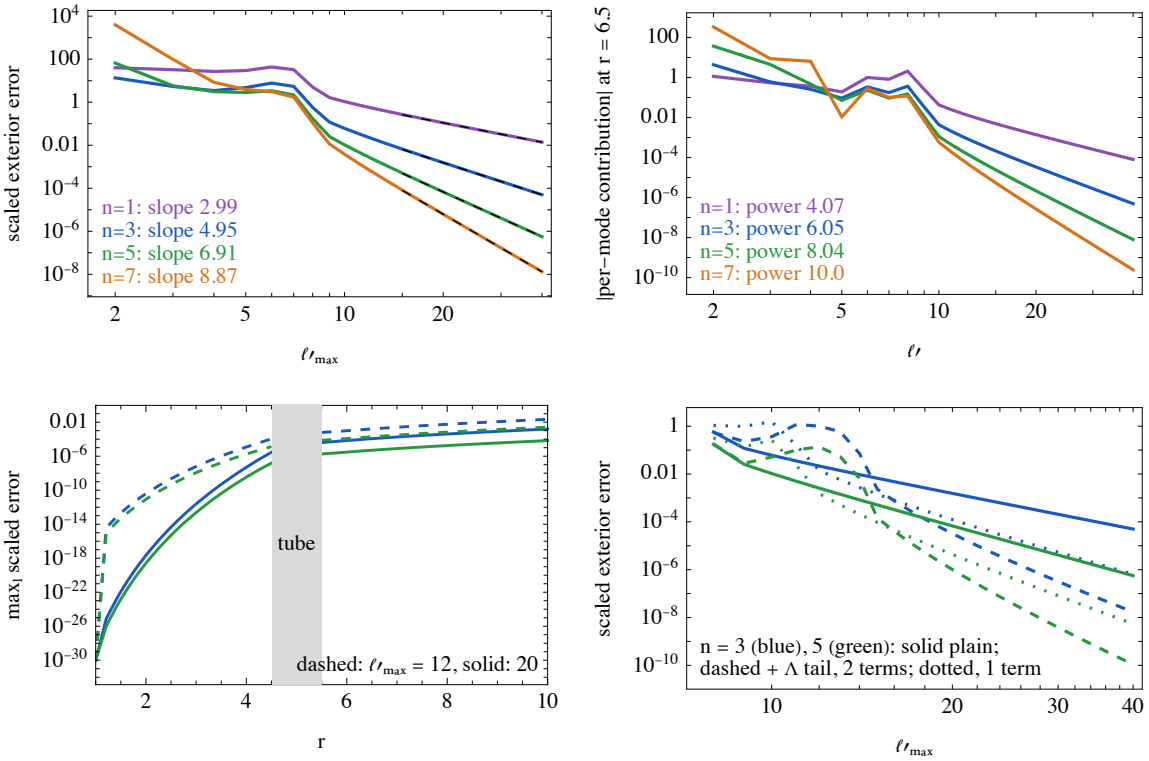


FIG. S3. Hybrid split, $Q = \frac{1}{2}\partial_z^2(AB)$, $k = 0$. Top: error against $\ell'_{\max}$ with fitted slopes, and per-mode contributions against ℓ'. Bottom left: error against r. Bottom right: tail acceleration.

S6. COMPARISON WITH MUSAEUS, POUND AND UPTON

Mode-level finite part. For the operator $r^2\partial_r^2 - 2r\partial_r - (\ell-1)(\ell+2)$ with the Green's function of their Eq. 140, and sources $A/\Delta r^4 + B/\Delta r^3 + C/\Delta r^2 + D/\Delta r$, we compared three calculations for $\ell = 2$–6:

- their prescription (Eqs. 141 and A2–A5);
- an independent Taylor-subtracted finite part, which agrees with it to below 10^{-38}, independently of the cutoff;
- the contour integral $\mathrm{Re}\int_C G\,S\,dz$, which agrees with both to 4×10^{-34} for five choices of worldtube radius and contour height.

Their closed form, Eq. 142, is confirmed.

T_Q and the mode-level δ content (exact puncture, $Q = \frac{1}{2}\partial_z^2(AB)$). The canonical SR is $q\,\partial_z^2(\Phi^{\mathrm{R}}/\rho)$, a derivative of an integrable function. Its 4D finite part is the ball-excluded principal value, and the difference between the two is $T_Q = -\frac{4\pi}{3} q\Phi^{\mathrm{R}}(\mathbf{x}_0)\delta^3$, verified to 10^{-21}.

At mode level, the total δ coefficient is $c_\ell = -4\pi q\Phi^{\mathrm{R}}(\mathbf{x}_0)Y_{\ell 0}(0)/r_0^2$. It splits as $\frac{2}{3}c_\ell$ from the finite part plus $\frac{1}{3}c_\ell$ from T_Q. Scaled exterior errors for each prescription:

prescription	error ($\ell \le 6$)
contour, canonical source	$< 10^{-27}$
modes of the 4D finite part + T_Q (their Eq. 24)	$< 10^{-17}$
modes of the 4D finite part, without T_Q	0.14–170
pointwise modes, mode-level finite part (their Eq. 105) + T_Q	0.27–340
pointwise modes, no δ terms	0.41–500

The shell (mode-level) and ball (4D) regulators do not commute for the trace-free ρ^{-3} term: a term proportional to $B'/(B+B')$ survives.

S7. NUMERICAL DETAILS

All computations use working precision 30 and exact rational parameters. NDSolve is run with PrecisionGoal 20, AccuracyGoal 35 and `InterpolationOrder -> All`. Contour sources are evaluated at numerical $y(a)$. Symbolic composition expands them into high-degree polynomials and loses all accuracy.

The Cauchy step uses Gauss quadrature after subtracting the cubic Taylor polynomial of s about $\mathrm{Re}\, z$, which is integrated analytically. The source on the contour is represented piecewise at 40-digit precision.

The full test suite has more than 2,000 automated checks.

[1] S. D. Upton, B. Wardell, A. Pound, N. Warburton, and L. Barack, arXiv:2508.00087 [gr-qc] (2026).
[2] A. Heffernan, A. Ottewill, and B. Wardell, Phys. Rev. D **86**, 104023 (2012).